

# **Collaborative Filtering Report**

## **1. Data Preparation**

- We used a rating dataset containing user ratings (user\_id, book\_id, ratings).
- Constructed a user-item interaction matrix where each row represents a user, each column a book, and the values are ratings.
- We constructed a user-item interaction matrix using the `pivot_table()` method,
- We built a matrix where:
  - Rows = users
  - Columns = books
  - Values = user ratings
- Missing values were filled with 0, indicating no interaction.
- Shape: (38 users × 100 books)  
Sparsity: Very high, with most users rating only a few books. Missing values filled with 0, indicating no interaction.

## **2. User-User Collaborative Filtering**

- Used the cosine similarity metric with KNN (k=6) to find similar users.
- For a target user, we retrieved books liked by similar users but not yet rated by the user.

### **Performance:**

- Worked only for users with multiple overlapping ratings.
- Due to sparsity, most users received no recommendations, as there were insufficient overlapping ratings between users.

### **Strength:**

- Works well when users have many overlapping ratings and when the ratings overlap strongly with other users.

### Weakness:

- Struggles with sparse data, where most users have rated very few books, leading to a lack of recommendations.

### Example Output:

```
User-User Recommendations for User 276726: ['8879839993', '0091830893', '0586207414', '0812571029', '8423996565']
User-User Recommendations for User 276729: ['0395547032', '8423996565', '8426449476', '8426449573', '8478884831']
User-User Recommendations for User 276736: ['8879839993', '0395547032', '8423996565', '8426449476', '8426449573']
User-User Recommendations for User 276737: ['8879839993', '0395547032', '8423996565', '8426449476', '8426449573']
User-User Recommendations for User 276744: ['0440414121', '8423996565', '8426449476', '8426449573', '8478884831']
```

## 3. Item-Item Collaborative Filtering

- Compares books by co-rating similarity.
- If a user liked Book A, other books rated similarly by other users are recommended.

### Strength:

- Performs well even if the user has rated only one book.
- Better than User-User CF for sparse data.
- Can recommend books to users with minimal interaction if the book they rated has strong similarity scores with others.

### Weakness:

- Relies on the quality of book similarity, which can be poor in cases of very sparse data.

### Example Output:

```
User 276726 => Item-Item Recommendations: ['0006379702', '3442422035', '3453213025', '3453137442', '3453092007']
User 276729 => Item-Item Recommendations: ['3442435773', '347354034X', '3453213025', '3453137442']
User 276736 => Item-Item Recommendations: ['3442422035', '3453213025', '3453137442', '3453092007', '3442449820']
User 276737 => Item-Item Recommendations: ['0006379702', '3442435773', '3453213025', '3453137442', '3453092007']
User 276744 => Item-Item Recommendations: ['0006379702', '3442435773', '3453213025', '3453137442', '3453092007']
```

## 4. Matrix Factorization (SVD)

- Applied TruncatedSVD (20 components) to reduce dimensionality.
- Reconstructed matrix to predict ratings for all books.

- Recommended top N books with highest predicted ratings.

### Performance:

- Generated high-coverage personalized recommendations for every user.
- Performed best in sparse environments due to its ability to generalize via latent dimensions.
- Best performance among all three techniques.

### Strength:

- Handles sparsity well by utilizing latent dimensions to predict missing ratings.
- Offers personalized and diverse recommendations.

### Weakness:

- SVD is computationally more complex compared to simpler methods like Item-Item CF.

### Example Output:

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```
User 276726 => SVD Recommendations: ['0671537458', '0060517794', '3499230933', '3596151465', '0679776818']
User 276729 => SVD Recommendations: ['3596218098', '0395547032', '8484332039', '8484330478', '342310538']
User 276736 => SVD Recommendations: ['3442136644', '0747558167', '0684867621', '0440414121', '3596218098']
User 276737 => SVD Recommendations: ['8879839993', '0440414121', '8484332039', '8484330478', '055310666X']
User 276744 => SVD Recommendations: ['0330332775', '8879839993', '1562827898', '8440682697', '0671537458']
```

## 5. Evaluation and Comparisons:

```
Recommendations for User 276726
Books already rated: ['0155061224']
User-User CF Recommendations: ['0345443683', '043935806X', '055310666X', '8484330478', '8484332039']
Item-Item CF Recommendations: ['0006379702', '8437606322', '3442136644', '3453092007', '3453213025']
SVD Recommendations: ['0671537458', '0060517794', '3499230933', '3596151465', '0679776818']
```

```
Recommendations for User 276729
Books already rated: ['052165615X', '0521795028']
User-User CF Recommendations: ['0345443683', '043935806X', '055310666X', '8484330478', '8484332039']
Item-Item CF Recommendations: ['052165615X', '8437606322', '3442136644', '3453092007', '3453213025']
SVD Recommendations: ['3596218098', '0395547032', '8484332039', '8484330478', '342310538']
```

```
Recommendations for User 276736
Books already rated: ['3257224281']
User-User CF Recommendations: ['0345443683', '043935806X', '055310666X', '0006379702', '3442131340']
Item-Item CF Recommendations: ['8426449573', '3442131340', '3442136644', '3453092007', '3453213025']
SVD Recommendations: ['3442136644', '0747558167', '0684867621', '0440414121', '3596218098']
```

## **Recommendations for User 276726**

Books already rated:

- '0155061224'

User-User CF Recommendations

- ['0345443683', '043935806X', '055310666X', '8484330478', '8484332039']
  - Explanation: User-User Collaborative Filtering found similar users based on overlapping ratings and recommended books that were liked by these similar users.

Item-Item CF Recommendations

- ['0006379702', '8437606322', '3442136644', '3453092007', '3453213025']
  - Explanation: Item-Item Collaborative Filtering identified books that share similar ratings patterns with those already rated by the user, providing new recommendations based on those similarities.

SVD Recommendations

- ['0671537458', '0060517794', '3499230933', '3596151465', '0679776818']
  - Explanation: Matrix Factorization (SVD) utilized latent features from the user-item interaction matrix to predict ratings for unseen books, offering personalized recommendations with better coverage and diversity.

## **Recommendations for User 276729**

Books already rated:

- '052165615X', '0521795028'

User-User CF Recommendations

- ['0345443683', '043935806X', '055310666X', '8484330478', '8484332039']
  - Explanation: User-User Collaborative Filtering leveraged the similarity of ratings between this user and others to suggest books that users with similar tastes have enjoyed.

Item-Item CF Recommendations

- ['052165615X', '8437606322', '3442136644', '3453092007', '3453213025']
  - Explanation: Based on the co-rating patterns of books, Item-Item Collaborative Filtering recommended items that are similar to those the user has already rated.

#### SVD Recommendations

- ['3596218098', '0395547032', '8484332039', '8484330478', '342310538']
  - Explanation: SVD provided diverse recommendations based on learned latent features from the interaction matrix, delivering highly personalized suggestions even for users with multiple ratings.

### **Recommendations for User 276736**

Books already rated:

- '3257224281'

#### User-User CF Recommendations

- ['0345443683', '043935806X', '055310666X', '0006379702', '3442131340']
  - Explanation: User-User Collaborative Filtering suggested books based on similar users' ratings, although the recommendations are constrained by the available overlap in ratings.

#### Item-Item CF Recommendations

- ['8426449573', '3442131340', '3442136644', '3453092007', '3453213025']
  - Explanation: Item-Item Collaborative Filtering identified books with similar ratings patterns to those the user rated, offering relevant suggestions based on item similarities.

#### SVD Recommendations

- ['3442136644', '0747558167', '0684867621', '0440414121', '3596218098']
  - Explanation: SVD delivered high-coverage recommendations by exploiting latent features, handling sparsity effectively and generating personalized suggestions even with limited ratings.

### **Metrics Evaluation :**

- **Mean Absolute Error (MAE):**

The average of absolute errors between predicted ratings and actual ratings is 0.038, which is very small and indicates that the system's predicted ratings are relatively accurate on average.

- **Root Mean Squared Error (RMSE):**

The RMSE of 0.447 suggests that the predictions have some error but are still in a reasonable range for recommendation systems.

- **Coverage:**

The coverage of 0.16 suggests that only 16% of all possible books have been recommended across users. A low coverage could indicate that the system is only recommending a small subset of books, which may be a limitation of the current model.

- **Diversity:**

A diversity score of 0.808 indicates that the system is recommending a diverse set of books, which is good for providing a variety of suggestions to users.

### **Insight:**

- SVD consistently provides better recommendations than both User-User and Item-Item CF, especially for users with limited data.
- SVD excels in coverage and reliability, offering the best results in sparse environments.

### **Observations**

- User-User CF struggles with sparse data: No overlapping ratings between users lead to no recommendations.
- Item-Item CF performs reliably: Even with minimal ratings, it can recommend books based on item similarity.
- Item-Item CF is more effective when data is sparse.
- SVD delivers the best overall recommendations: Handles sparse user data effectively and provides consistent, broad suggestions by uncovering latent patterns in the data.